Day 19: Odds, odds ratios, and logistic regression

> **Note**
>
> These are reference notes for Day 19. They cover the same material as the Day 19 slides and the self-paced tutorial.

Where we're going

So far, our response variable has always been **quantitative**. Today the response is **binary** (yes/no, success/failure), so we need a new model: **logistic regression**.

Data: fake news

150 news articles, some real and some fake. Our goal is to explain `type` using features of the article.

- `type`: "real" or "fake"; `type_num`: 0 (real) or 1 (fake)
- `title_words`: number of words in the title
- `title_has_excl`: whether the title has an exclamation point
- `title_caps_percent`: percent of the title in capital letters

A "success" ($Y = 1$) is a **fake** article. Always check how your response is coded before modeling!

Probability and odds

If π is the probability of an event, the **odds** of the event are

$$\text{odds} = \frac{\pi}{1-\pi} = \frac{P(\text{event})}{P(\text{not event})} \qquad \pi = \frac{\text{odds}}{1+\text{odds}}$$

Probability π	Odds $\pi/(1-\pi)$	How we'd say it
0.5	1	"even odds"
0.75	3	"3 to 1"
0.2	0.25	"1 to 4"

Probabilities are between 0 and 1; odds are between 0 and ∞.

Example: exclamation points

	fake	real	Sum
FALSE	44	88	132
TRUE	16	2	18
Sum	60	90	150

	Title has exclamation	No exclamation
$P(\text{fake})$	$16/18 = 0.889$	$44/132 = 0.333$
$\text{Odds}(\text{fake})$	$16/2 = 8$	$44/88 = 0.5$

Shortcut: from a table, odds = (number of successes) / (number of failures).

The **odds ratio** compares the odds in two groups:

$$OR = \frac{\text{Odds}(\text{fake} \mid \text{exclamation})}{\text{Odds}(\text{fake} \mid \text{no exclamation})} = \frac{8}{0.5} = 16$$

Interpretation: The odds of being fake are 16 times as high for articles with an exclamation point in the title as for articles without one.

> ⚠ Common mistake
>
> An odds ratio is **not** a ratio of probabilities. It is *not* correct to say that articles with an exclamation point are "16 times as likely" to be fake: the ratio of probabilities is $0.889/0.333 = 2.67$.

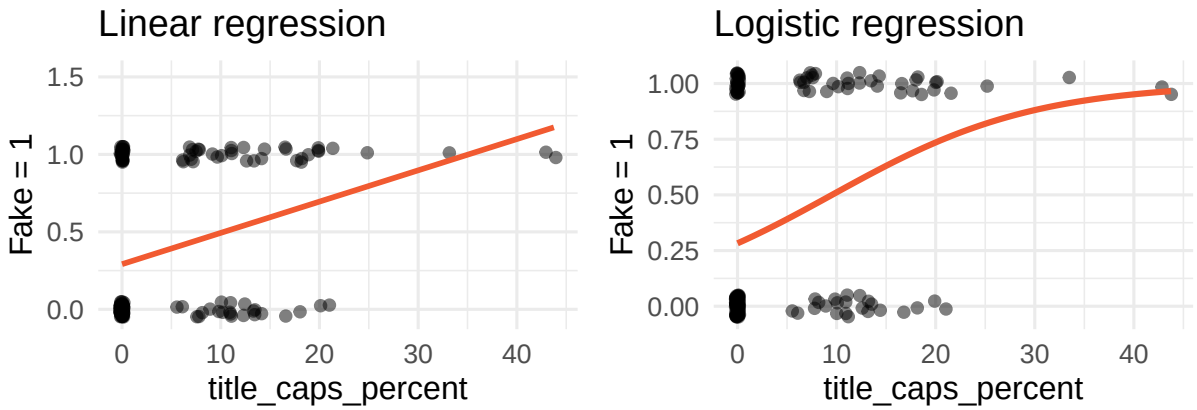
Q1. An odds ratio of exactly 1 means _____

_____.

An odds ratio less than 1 means _____

_____.

Why not linear regression?



A line keeps going forever, so it predicts "probabilities" outside of $[0, 1]$: the linear model predicts 1.17 for the title with the most capital letters (43.75%). Logistic regression gives an S-shaped curve that stays between 0 and 1.

The logistic regression model

$$\log\left(\frac{\pi(x)}{1-\pi(x)}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p$$

where $\pi(x) = P(Y = 1 \mid x)$. The left side is called the **log-odds** or the **logit**. It can be any number, so it's OK for the right side to be a line.

Scale	Formula	Range	Linear in the x's?
Probability	π	0 to 1	No
Odds	$\pi/(1-\pi)$	0 to ∞	No (exponential)
Log-odds	$\log(\pi/(1-\pi))$	$-\infty$ to ∞	**Yes**

Fitting the model in R

```
news_glm <- glm(type_num ~ title_caps_percent, data = news, family = binomial)
```

term	estimate	std. error	t	p-value
Intercept	-0.932	0.220	-4.24	<0.001
title_caps_percent	0.098	0.025	3.84	<0.001

$$\hat{\eta} = \log\left(\frac{\hat{\pi}(x)}{1-\hat{\pi}(x)}\right) = -0.932 + 0.098x$$

where x is `title_caps_percent`. (The slides rounded these to -0.93 and 0.1, so some numbers there are slightly different.)

Interpreting the slope

$\hat{\beta}_1 = 0.098$ is the change in the **log-odds** for a 1-unit increase in x. To get to the odds scale, exponentiate:

$$\widehat{\text{odds}}(x) = e^{\hat{\beta}_0 + \hat{\beta}_1 x} \quad \Longrightarrow \quad \frac{\widehat{\text{odds}}(x+1)}{\widehat{\text{odds}}(x)} = \frac{e^{\hat{\beta}_0} e^{\hat{\beta}_1 (x+1)}}{e^{\hat{\beta}_0} e^{\hat{\beta}_1 x}} = e^{\hat{\beta}_1}$$

So $e^{\hat{\beta}_1}$ is an **odds ratio**. Here, $e^{0.098} \approx 1.103$:

> For each additional percentage point of capital letters in the title, the odds of the article being fake are multiplied by about 1.10 (an increase of about 10%).

For a c-unit increase in x, the odds are multiplied by $e^{c\hat{\beta}_1}$. For example, 10 more percentage points of capital letters multiplies the odds by $e^{10(0.098)} \approx 2.66$ (**not** 10×1.10).

> ⚠ Common mistakes
>
> - $\hat{\beta}_1$ is **not** the change in the probability. In logistic regression, the change in probability for a 1-unit increase in x depends on x.
> - $e^{\hat{\beta}_1}$ **multiplies** the odds; it is not added to them.
> - $e^{\hat{\beta}_1}$ multiplies the **odds**, not the probability.

From log-odds to probability

What's the estimated probability that an article with **20%** capital letters in the title is fake? Work from the inside out:

1. **Log-odds:** $\hat{\eta} = -0.932 + 0.098(20) = 1.023$
2. **Odds:** $\hat{\omega} = e^{\hat{\eta}} = e^{1.023} = 2.78$
3. **Probability:** $\hat{\pi} = \dfrac{\hat{\omega}}{1+\hat{\omega}} = \dfrac{e^{\hat{\eta}}}{1+e^{\hat{\eta}}} = \dfrac{2.78}{1+2.78} = 0.735$

The intercept works the same way: at $x = 0$, the log-odds are -0.932, the odds are $e^{-0.932} = 0.394$, and the probability is 0.283.

In R, `predict()` does this for you, but by default it gives the **log-odds**. Use `type = "response"` to get probabilities:

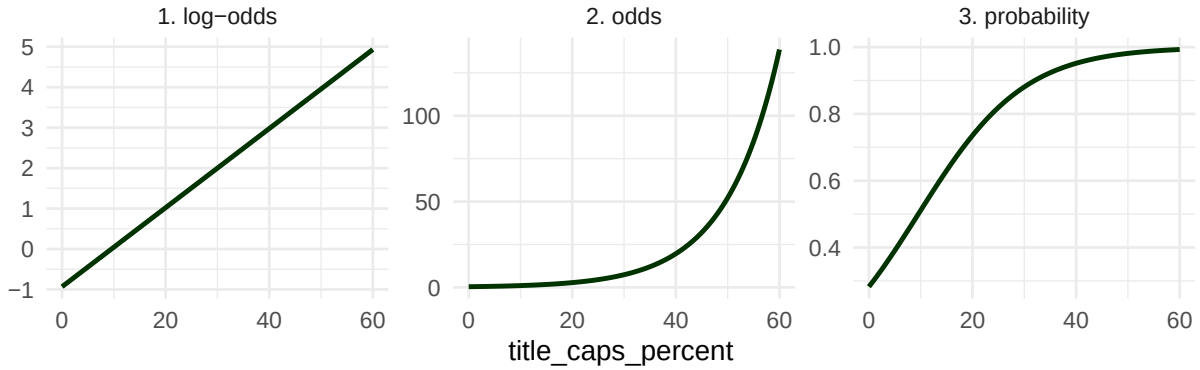
```
predict(news_glm, newdata = data.frame(title_caps_percent = 20))
```

```
       1
1.022656
```

```r
predict(news_glm, newdata = data.frame(title_caps_percent = 20), type = "response")
```

```
        1
0.7354896
```

The fitted model on three scales



Q2. On the log-odds scale, the fitted model is a _____. On the odds scale, it is a(n) _____ curve. On the probability scale, it is a(n) _____ curve.

The probability curve flattens out near 1 because _____

_____.

Moving between the three scales

From	To	Formula	In R
log-odds η	odds	e^{η}	`exp(eta)`
odds	probability	$\frac{\text{odds}}{1+\text{odds}}$	`odds / (1 + odds)`
log-odds η	probability	$\frac{e^{\eta}}{1+e^{\eta}}$	`exp(eta) / (1 + exp(eta))`
probability π	odds	$\frac{\pi}{1-\pi}$	`p / (1 - p)`
odds	log-odds	$\log(\text{odds})$	`log(odds)`

A binary predictor gives back the odds ratio

```
excl_glm <- glm(type_num ~ title_has_excl, data = news, family = binomial)
exp(coef(excl_glm))
```

```
      (Intercept) title_has_exclTRUE
              0.5               16.0
```

- $e^{\widehat{\beta}_1} = 16$ is exactly the odds ratio we computed from the table.

Q3. What does $e^{\widehat{\beta}_0} = 0.5$ represent in context?

Your turn: three predictors

Fit `glm(type_num ~ title_has_excl + title_caps_percent + title_words, data = news, family = binomial)`.

Q4. Write out the fitted model on the log-odds scale for articles with and without an exclamation point in the title.

(A) With an exclamation point:

$$\log\left(\frac{\hat{\pi}}{1-\hat{\pi}}\right) = _____ + _____ \times \texttt{title_caps_percent} + _____ \times \texttt{title_words}$$

(B) Without an exclamation point:

$$\log\left(\frac{\hat{\pi}}{1-\hat{\pi}}\right) = _____ + _____ \times \texttt{title_caps_percent} + _____ \times \texttt{title_words}$$

Q5. Interpret each slope coefficient in terms of the **odds** of being fake news.

`title_has_excl`:

`title_caps_percent`:

`title_words`:

Q6. The estimated probability that an article with an exclamation point, 10% capital letters, and 12 words in the title is fake: _____

Q7. Which predictors appear to be most important for this model? Why?

Q8. What questions do you have about the `summary()` output?

Interpretation templates

Quantity	Interpretation
$\widehat{\beta}_j$	Change in the log-odds of success for a 1-unit increase in x_j, holding other predictors constant
$e^{\widehat{\beta}_j}$, quantitative x_j	For a 1-unit increase in x_j, the odds of success are multiplied by $e^{\widehat{\beta}_j}$, holding other predictors constant
$e^{\widehat{\beta}_j}$, indicator x_j	The odds of success are $e^{\widehat{\beta}_j}$ times as high for the indicated group as for the reference group, holding other predictors constant
$e^{\widehat{\beta}_0}$	The odds of success when all predictors equal 0 (or for the reference group)

R function reference

Function	Purpose
`table(x, y) |> addmargins()`	Two-way table with row and column totals
`glm(y ~ x, data, family = binomial)`	Fit a binary logistic regression model
`coef(model)`, `exp(coef(model))`	Coefficients on the log-odds scale; odds ratios
`summary(model)`	Model summary and Wald tests
`predict(model, newdata)`	Predictions on the log-odds scale
`predict(model, newdata, type = "response")`	Predictions on the probability scale